

Application form mySNF

Instrument **Scientific Exchanges**

Part 1: General Information

Basic data

Project Title	Biodiversity Dynamics in Coevolutionary Metaecosystems
Project title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems

Research Field	Life sciences
Main Discipline	30207 Ecology
University	Eidg. Anstalt für Wasserversorgung - EAWAG

Applicant(s)

Main Applicant	Carlos Melian
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Grant Application

Amount requested (CHF)	Total	13'500
Requested starting date	01.10.2018	
Duration (months)	1	

Attachments

CV and publication list	CVs_All_Invited_Participants.pdf CV_CarlosMelian.pdf
Further documents for scientific events or research visits	Form_confirmation_self_declaration_Signed.pdf EventProgramme.pdf
Other annexes	ProposalfortheWorkshop.pdf

1. Responsible applicant

Last name	Melian
First name	Carlos
Function (title)	Group Leader
Academic degree	Dr. / PhD
Date of birth	23.08.1969
Gender	male
Civil Status	Married
Language	German
Nationality	Spain
Correspondence address of application	Address of workplace

Home address

Address supplement	
Street, No.	Seestrasse 79
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
Country	Switzerland

Address of institute

Name of Institution 1 (e.g. laboratory) *	Eawag
Continuation 1 (e.g. inst / dept.)	
Continuation 2 (e.g. University)	
Street, No.	Seestrasse 79
Address supplement 1 (e.g. building)	
Address supplement 2 (e.g. office)	
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
State, canton, etc.	
Country	Switzerland

Communication

Secretariat line	
Switchboard	
Direct line	+41 58 765 22 08
Fax office	
Home telephone number	
Cellphone	+41 78 722 88 73
Website	http://www.eawag.ch/about/personen/homepages/melianca/index
E-mail address	carlos.melian@eawag.ch

2. Applicants' employment

Information on employment and function at the anticipated starting date of the grant

Name	Melian, Carlos
Employment at the anticipated starting date of the grant	permanent contract
Level of employment %	100
Function in the context of this grant application	Head of (e.g. institute, department, center, clinic)
Professorship	Assistant professor with tenure track
Doctorate (PhD)?	Yes
Date of doctorate (PhD)	01.02.2005
PhD supervisor	
Country of doctorate	Spain
Remarks	
Further employments	

3. Basic data I

Original title	Biodiversity Dynamics in Coevolutionary Metaecosystems
Title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems
Requested starting date	01.10.2018
Duration (months)	1
Research field	Life sciences
Main discipline	30207 Ecology
Sub-discipline(s)	30206 Environmental Research

4. Basic data II

Summary

Background and Rationale
Joining theoretical and empirical evolutionary ecologists to develop and test theoretical models that incorporate the empirical patterns of reciprocal trait changes will be critical to improve our predictions on biodiversity dynamics and their responses to multiple disturbances in species- and interaction-rich ecological communities. The aim of the workshop is to integrate trait-species-space interactions data into new process-based coevolutionary metaecosystem models. Such integration might facilitate the exploration of patterns of coexistence of co-evolved species, their role in ecosystem functioning, and the risk of collapse of co-evolved ecosystems across different spatio-temporal scales, which is of critical importance for three reasons. First, the extinction of species caused by human-generated disturbances occurs at an unprecedented rate, and yet most models predicting extinction rates often assume that species evolve independently of one another. Second, although eco-evolutionary approaches of species interactions are increasingly common, studies connecting co-evolving trait distributions within and between species at small scales to investigate the large-scale association between coevolutionary hot spots, biodiversity function and meta-ecosystem dynamics are rare. Third, although many experiments and empirical studies have shown convergence of time scales between ecological and coevolutionary processes, there is a gap in understanding how ecological and coevolutionary dynamics in species-rich assemblages shape biodiversity and ecosystem function in a spatio-temporal context.

Objectives and Methods
We will focus on implementing analytic tools and testing them with empirical patterns from data sets containing many species, sites and traits. We propose to bring together biologists skilled in theoretical ecology, evolutionary biology, and community ecology to discuss about the following two questions:
Question 1: How does reciprocal trait change influence biodiversity dynamics, species extinctions and ecosystem function in species-rich meta-ecosystems?
We aim to integrate the effects of ecological, coevolutionary, and migration dynamics in shaping biodiversity in meta-ecosystems. We will merge the meta-ecosystem perspective into existing models of coevolutionary interaction networks. We will implement scenarios to predict biodiversity distribution across space and time under different functional interactions (i.e., trait matching and exploitation barriers, assuming weak and strong reciprocal selection strengths. Metacommunity and meta-ecosystem theory approaches will complement the functional scenarios to understand how spatial processes governed by dispersal or niche width can alter the coevolutionary dynamics within communities by contributing to species trait variation and network structure and stability, and thus ecosystem functioning.
Question 2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems?
Most interactions between species generate reciprocal selection pressures, and the coevolutionary dynamics that emerges over time varies with the fitness consequences of the ecological interaction (i.e., competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions between species. Yet, how the relationship between species traits and coevolution at local scales and the arrangement of ecosystems at the landscape scale produce rapid biodiversity changes and coevolutionary hot spots is not well understood. We will run simulations under landscapes with different spatial arrangements and migration scenarios to disentangle the interplay between coevolutionary dynamics at the local and the landscape level.

Keywords	Expected results and Impact We see great potential in linking the research of the participants to produce new synthesis in merging coevolution, interaction networks and meta-ecosystems. Specifically, the workshop aims to decipher the interplay between coevolution in local ecosystems for antagonistic and mutualistic interactions with the effects of migration under different landscape arrangements. This is a venue that can move forward the needed synthesis between the ecological and evolutionary processes that shape biodiversity. Furthermore, our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function is just starting to emerge. Yet, the integration between robust empirical patterns and process-based models, the main aim of this workshop, remain at a very incipient stage.
	Biodiversity dynamics
	Metacommunity theory
	Coevolution
	Theoretical ecology
	Biodiversity patterns
Language of correspondence	German
Financial administration	EAWAG Verwaltung zHv Frau Barbara Breu Rütli

5. University or research institution

University	Eidg. Anstalt für Wasserversorgung - EAWAG
Remarks	

6. Requested funding

Requested funding	Total (CHF)	Year 1
Total (CHF)	13'500	13'500

Research funds	Total (CHF)	Year 1
Travel & subsistence allowances	13'500	13'500
Total (CHF)	13'500	13'500
Total (%)	100%	100%

Details

Travel & subsistence allowances	Total (CHF)	Year 1
Andreazzi, Cecilia, Associate Researcher, Fundação Oswaldo Cruz, Rio de Janeiro, Brasil	2'360	2'360
Relation to research plan / Comments / Additions	Airfare (Brasil) 1750CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF	
Arim, Matias, Full Professor, University of the Republic, Maldonado, Uruguay	2'380	2'380
Relation to research plan / Comments / Additions	Airfare (Uruguay) 1770CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF	
Astegiano, Julia, Researcher, CONICET, Cordoba, Argentina	2'360	2'360
Relation to research plan / Comments / Additions	Airfare (Argentina) 1750CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF	

	Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Becks, Lutz, Researcher, Max-Planck Inst., Plön, Germany		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (Germany) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Heleno, Ruben, Auxiliary Researcher, University of Coimbra, Portugal		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (Portugal) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Huang, Weini, Lecturer, Queen Mary Univ. London, London, UK		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (UK) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Massol, François, Researcher, CNRS, Lille, France		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (France) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Sabatino, Malena, Researcher CONICET, Mar del Plata, Argentina		2'360	2'360
Relation to research plan / Comments / Additions	Airfare (Argentina) 1750CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Total (CHF)		13'500	13'500
Total (%)		100%	100%

7. Timeframe of the scientific exchange

Starting date	15.10.2018
Place	Kastanienbaum, Switzerland
Duration (days)	5

8. Contents

General information

Aim and motivation

Motivation and AimCoevolution, i.e. changes of species traits driven by reciprocal selection within and among interacting species, has been invoked as a primary mechanism generating much of Earth's biodiversity. Two main theories have been proposed to explain the dynamics of coevolutionary diversification. The Fisherian sexual selection theory, which states that female and male traits can lead to sexual isolation and speciation, and the Geographic mosaic theory of coevolution, which suggests that reciprocal changes in traits determining species interaction produce coevolutionary diversification as interspecific interactions control reproductive isolation and speciation. Despite the potential of such theories for explaining much of Earth's biodiversity, understanding how coevolution of intra- and inter-specific traits accounting for ecological interactions alter ecosystem dynamics across heterogeneous

landscapes remain elusive. Such understanding is particularly important in ecosystems with species- and interaction-rich ecological networks because global changes might trigger many species extinctions and the collapse of ecosystems. The aim of the workshop is to integrate trait-species-space interactions data into new process-based coevolutionary meta-ecosystem models to understand the impact of coevolution on the dynamics of biodiversity. We will focus on implementing analytic tools and testing them with empirical patterns from data sets containing many species, sites and traits. We propose to bring together biologists skilled in theoretical ecology, evolutionary biology, and community ecology to discuss about the following two questions: 1: How does reciprocal trait change influence biodiversity dynamics, species extinctions and ecosystem function in species-rich meta-ecosystems? 2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems? Added value for Swiss-research The workshop will be a one-off event, and it will lead to new international collaborations between Swiss institutions and institutions from abroad. One of the key milestones of the workshop will be to draft new ideas to produce a grant proposal draft to be submitted to the H2020, Biodiversa, and/or sDiv working groups. We also see great potential in linking the research of the participants based in Switzerland to produce new synthesis in merging coevolution, interaction networks and meta-ecosystems. Specifically, deciphering the interplay between coevolution in local ecosystems for antagonistic and mutualistic interactions with the effects of migration under different landscape arrangements is a venue that can move forward the needed synthesis between the ecological and evolutionary processes that shape and maintain biodiversity. Furthermore, our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function is just starting to emerge. Yet, the integration between robust empirical patterns and process-based models, the main aim of this workshop, is currently missing. Expected results and communication of results We will work on an opinion, review or ideas-perspectives paper entitled "Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems" for submission to Trends in Ecology and Evolution, Ecology Letters or Biological reviews. Prior to the meeting, attendees will provide written perspectives about the main overarching questions in their fields and the trait-species-spatial interaction data they can provide. On the first day, participants will present these ideas based on their own research programs. On subsequent days we will synthesize these perspectives into a single review/opinion and synthesis style paper integrating the datasets to the methods that we will develop and implement during the workshop and following the planned research visit.

Organisation

The workshop will be hosted at Eawag's Center for Ecology, Evolution & Biogeochemistry (CEEB), in Kastanienbaum, Switzerland. This is an ideal venue because all the invited guests can stay on-site in Eawag apartments, directly on the shore of Lake Lucerne. The meeting facilities are very suitable and have been previously used to host several small workshops and student courses with great success. Such events also help to raise our international profile, and, in the past, have led to repeat visits from graduate students and visiting faculty on sabbatical terms. Eawag will provide the necessary infrastructure for the workshop (venue, consumables, IT, etc.). No other funds are needed to support this workshop.

Target audience

Expected number of people (total number of participants)

14

Remark

We have invited 8 scientists from abroad (7 countries) and 6 from Switzerland working in different departments and institutions (Eawag, Ecology and Fish Ecology and Evolution department) and WSL.

Active participants

Number of active participants from the academic field

7

Number of active participants from outside

7

the academic field

Remark

We have selected a balanced group of participants representing a wide range of scientific ages, expertise, and levels of experience from the academic field object of the proposal (7) and outside the academic field (7). We have identified a mixture of people with (i) skills in mathematical biology, (ii) experience with software development, and (iii) strong conceptual skills in evolutionary biology, meta-ecosystem theory, model systems and biodiversity patterns from field studies.

Number of women

7

Number of PhD students

1

Number of students

0

Remark

We have carefully chosen on behalf of gender equality (7F:7M, from 7 countries), to represent diverse fields (e.g. theoretical ecologists, evolutionary biologists, and community ecologists) and scientific career stages (2 Profs., 5 Associate Prof. or group leaders (2 tenure track), 4 research scientists, 2 postdoctoral researches, and 1 PhD student).

9. General remarks on the project

Subject

Submitted in parallel with a research visit

Communication

This scientific event will be joined to a research visit submitted in parallel.

Confidential

No